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Further paleomagnetic results from the ~155 Ma Tiaojishan Formation, Yanshan Belt, North China, and their implications for the tectonic evolution of the Mongol–Okhotsk suture

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ABSTRACT

A new paleomagnetic study on well-dated (~155 Ma) volcanic rocks of the Tiaojishan Formation (Fm) in the northern margin of the North China Block (NCB) has been carried out. A total of 194 samples were collected from 26 sites in the Yanshan Belt areas of Luanping, Beipiao, and Shouwangfen. All samples were subjected to stepwise thermal demagnetization. After removal of a recent geomagnetic field viscous component, a stable high temperature component (HTC) was isolated. The inclinations of our new data are significantly steeper than those previously published from the Tiaojishan Fm in the Chengde area (Pei et al., 2011, Tectonophysics, 510, 370–380). Our analyses demonstrate that the paleomagnetic directions obtained from each sampled area were strongly biased by paleosecular variation (PSV), but the PSV can be averaged out by combining all the virtual geomagnetic poles (VGPs) from the Tiaojishan Fm in the region. The mean pole at 69.6°N/203.0°E ($A_{95} = 5.6°$) passes a reversal test and regional tilting test at 95% confidence and is thus considered as a primary paleomagnetic record. This newly determined pole of the Tiaojishan Fm is consistent with available Late Jurassic poles from red-beds in the southern part of the NCB, but they are incompatible with coeval poles of Siberia and the reference pole of Eurasia, indicating that convergence between Siberia and the NCB had not yet ended by ~155 Ma. Our calculation shows a ~1600-km latitudinal plate movement and crustal shortening between the Siberia and NCB after ~155 Ma. In addition, no significant vertical axis rotation was found either between our sampled areas or between the Yanshan Belt and the major part of the NCB after ~155 Ma.

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1. Introduction

The collisional Mongol–Okhotsk suture (MOS) which extends westward from the Udsky Gulf of the Okhotsk Sea to central Mongolia (Fig. 1) is widely accepted as an important tectonic boundary between the Mongolia–North China Block (MOB–NCB) and the Siberian component of a stable Eurasian continent (Zonenshain et al., 1990; Xu et al., 1997; Zorin, 1999; Cogné et al., 2005). The age and tectonic evolution of this suture has long been a subject of much debate. Most researchers agree that the suture formed by the progressive closure of the Mongol–Okhotsk Ocean from west to east in a 'scissor-like' manner (Zhao et al., 1990; Zonenshain et al., 1990; Kravchinsky et al., 2002; Tomurtogoo et al., 2005; Metelkin et al., 2010). However, the reported timing of final closure has ranged from Triassic to the Early Cretaceous (Zonenshain et al., 1990; Maruyama et al., 1997; Halim et al., 1998; Zorin, 1999; Parfenov et al., 2001; Tomurtogoo et al., 2005).

Paleomagnetism remains the most powerful tool in studying the convergence process between two continental blocks. Comparing paleomagnetic results from coeval strata on both sides of the Mongol–Okhotsk suture is critical to understanding its convergence history (Cogné et al., 2005; Metelkin et al., 2007a). With respect to the MOS, two concordant and well-dated Late Jurassic (~155 Ma) paleomagnetic poles have been published from the Siberian side of the suture (Kravchinsky et al., 2002; Metelkin et al., 2007a, 2010). In contrast, previously available Mesozoic data from the southern side of the MOS are contradictory to each other. The likely reason for this is that most Jurassic strata are not well-dated (Lin et al., 1985; Fang et al., 1988; Zhao et al., 1990; Gilder and Courtillot, 1997; Gilder et al., 1999). Recently, a new Late Jurassic paleomagnetic pole was obtained from ~155 Ma volcanic rocks of the Tiaojishan Formation (Fm) in the Chengde basin (Pei et al., 2011), north China (Fig. 1), thus providing an opportunity for comparing this pole to the coeval Siberian poles (Kravchinsky et al., 2002; Metelkin et al., 2007a). However, the Tiaojishan paleopole indicates an abnormally lower paleolatitude than other Late Jurassic poles from the southern part of the NCB (Gilder

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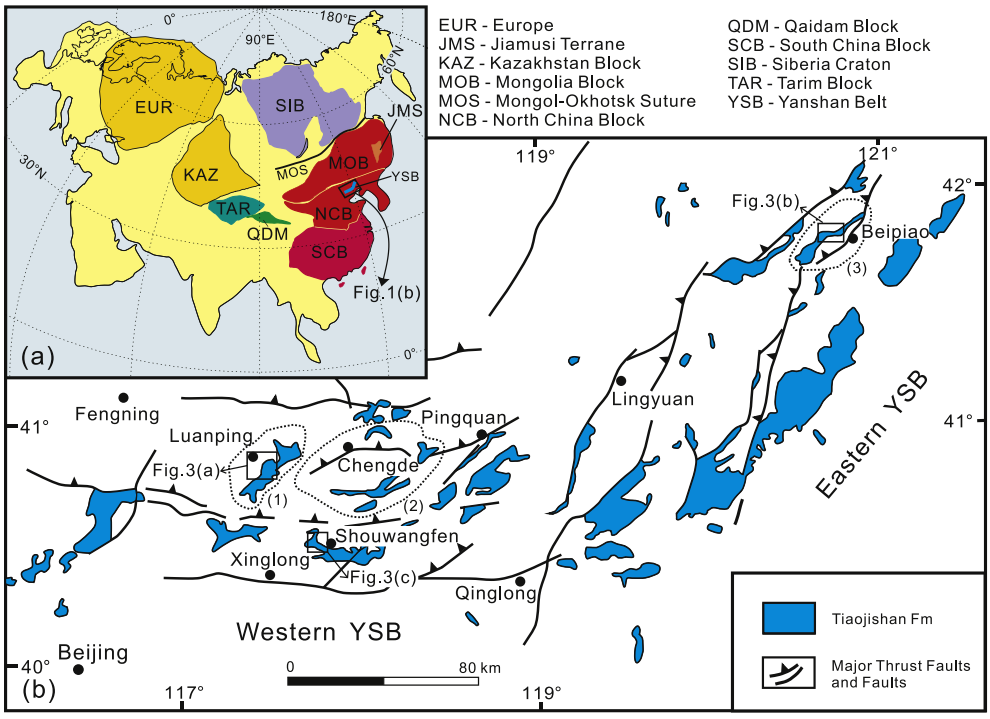


Fig. 1. (a) Tectonic setting of the Yanshan Belt (YSB) and the North China Block (NCB). (b) Distribution of the Tiaojishan Formation in the YSB, northern margin of the NCB, after Liu et al. (2006). Black dot lines are boundaries of the (1) Luanping basin, (2) Chengde basin, and (3) Beipiao basin.

and Courtillot, 1997; Gilder et al., 1999). More data are needed to test whether the Chengde basin pole could average out the paleosecular variation (PSV) of the paleogeomagnetic field, or whether the sampled section was affected by local vertical axis rotations in the Yanshan fold-thrust belt (YSB; Davis et al., 2001).

In this paper, we report new paleomagnetic results from four additional well-dated (~155 Ma) or well-correlated sections of the Tiaojishan Fm in the YSB (Fig. 1). Based on this new data, we discuss the PSV and local rotation issues, and the Late Jurassic–Cretaceous paleogeographic evolution of major crustal blocks in NE Asia.

2. Geological setting and paleomagnetic sampling

The YSB is a crustal deformation belt within the northern margin of the NCB (Fig. 1). In this region, Archean and Paleoproterozoic metamorphic rocks considered to be exposures of the cratonal basement are unconformably overlain by thick Mesoproterozoic to Neoproterozoic and Paleozoic sedimentary successions. They, in turn, are overlain unconformably by Mesozoic and Cenozoic terrestrial volcanic and clastic strata that were deposited in separate commonly fault-bounded basins (Fig. 1b). During an Early Jurassic through to Middle Cretaceous time interval, the YSB experienced major episodes of compressional and extensional deformations, which resulted in extensive development of folds, thrusts, normal faults, and shear zones (Zhao, 1990; Davis et al., 2001, 2009; Wang et al., 2010, 2013; Zhang et al., 2014b).

The Upper Jurassic Tiaojishan Fm is widely exposed in the YSB (Fig. 1). The thickness of the Tiaojishan Fm varies from basin to basin (~216 to 1953 m), and its fresh volcanic rocks are amenable to paleomagnetic study. The Tiaojishan Fm is composed of diverse volcanic rocks (basalt, andesite, dacite, rhyolite, tuff) and sedimentary rocks bearing plentiful plant fossils. It is conformably or disconformably underlain by the Jiulongshan (=Haifanggou) Fm and is overlain conformably by the Houcheng (=Tuchengzi) Fm (Bureau of Geology and Mineral Resources of Hebei Province, 1989; Bureau of Geology and Mineral Resources of Liaoning Province, 1989; Liu et al., 2006). The Jiulongshan and Houcheng Formations are mostly clastic rocks and have been dated, respectively, at 177.8 ± 7.7 to 161.6 ± 1.6 Ma by K/Ar whole rock analysis (Chen and Chen, 1997) and at 153.7 ± 1.1 to 137.4 ± 1.3 Ma, Zircon U-Pb ages of tuff beds (Zhang et al., 2008b; Xu et al., 2012; Fig. 2).

Recently, several U–Pb and $^{40}Ar/^{39}Ar$ ages of Tiaojishan volcanic rocks in the Luanping and Beipiao basins have been reported (Fig. 2; Fig. 3; Zhang et al., 2005, 2008a; Chang et al., 2009). An andesite flow near the top of the Tiaojishan Fm has been dated at 153.8 ± 5.2 Ma in Changshanyu of the Luanping basin, and a rhyolite dacite sample in the upper part of the formation near Beipiao city in Beipiao basin is dated at 154.0 ± 4.7 Ma; both age determinations by zircon U–Pb LA-ICP-MS method (Zhang et al., 2008a). Two precise $^{40}Ar/^{39}Ar$ dates (160.7 ± 0.4 Ma and 158.7 ± 0.6 Ma) for tuff-beds from the lowest part of the Tiaojishan Fm near Beipiao city were reported by Chang et al. (2009). Thus, the best estimated age for the Tiaojishan volcanic rocks was ca. 155 Ma (Pei et al., 2011).

Yanshan folds and thrusts trend roughly E–W in the western areas, whereas similar structures trend roughly NE in the eastern areas (Fig. 1b). It would be of wide interest to ascertain whether the eastern areas of the YSB have been rotated relatively to western areas after ~155 Ma (Wang, 1996; Davis et al., 2001; Liu et al., 2007; Wang et al., 2011b; Zhang et al., 2014b). Our sampled sections were, thus, collected in both western (Luanping and Shouwangfen basins) and eastern areas (Beipiao basin) of the YSB (Fig. 1b; Fig. 3).

A total of 194 core samples from 26 sites were collected from the Tiaojishan volcanic rocks in the Luanping basin (12 sites), Shouwangfen basin (3 sites), and Beipiao basin (11 sites). Volcanic and clastic rocks are interlayered in the sampled section in Luanping basin. The attitudes of the strata were measured on the beds of sandstone intercalated with the sampled volcanic rocks. In Shouwangfen basin, we sampled volcanic rocks at the top of the Tiaojishan Fm, which is marked by the boundary between the Tiaojishan and Houcheng Fms. The attitudes of the strata were also measured on the clastic interbed. In the Beipiao basin, we sampled two sections that are ~6 km apart; section I containing sites

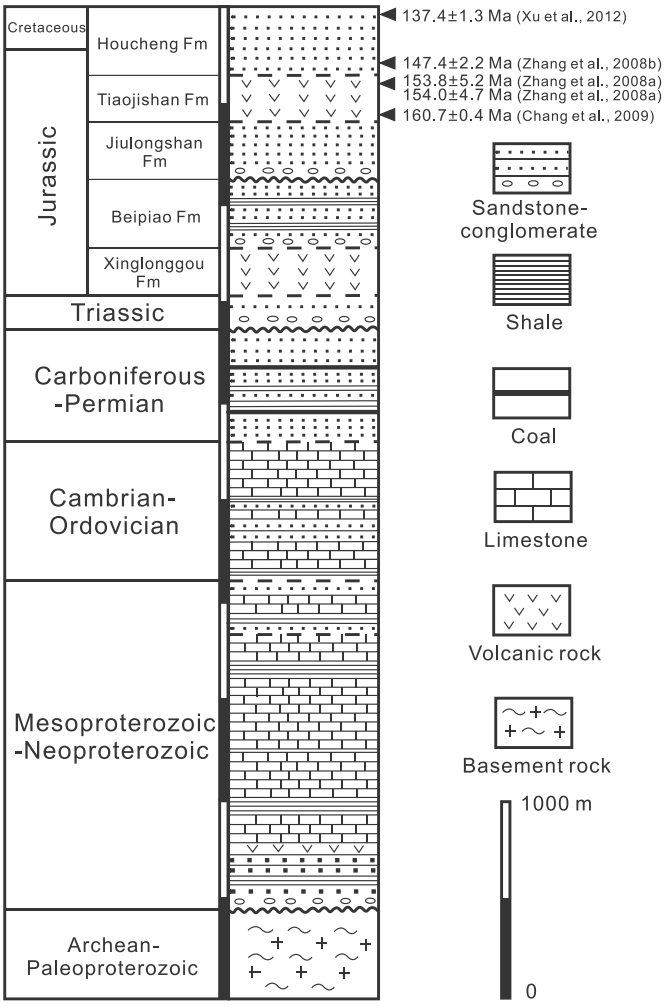


Fig. 2. Stratigraphic sequence of the YSB, North China Block.

BP01–06 and section II containing sites BP07–11. The attitudes of the strata were measured in the crystal tuff marker interbed. Generally, at each sampling site, five to 12 cores throughout several meters of stratigraphic thickness were sampled. All the samples were drilled with a water-cooled portable driller and oriented using magnetic and solar compasses. There were no significant orientation differences between the two compasses.

3. Laboratory techniques

Core samples were cut into 2.2-cm long specimens in the Paleomagnetism and Environmental Magnetism Laboratory of the China University of Geosciences, Beijing (CUGB). All specimens were subjected to stepwise thermal demagnetization up to 590 °C or 680 °C. The temperature intervals ranged generally from a maximum of 80 °C for lower temperature steps to a minimum of 5 °C for higher temperature steps. The remanent magnetization was measured using a 2G-755-4K cryogenic magnetometer. All rock magnetic measurements were performed within a μ-metal shielded room at CUGB with residual fields less than 200 nT. Remanent magnetization directions of all the specimens were analyzed using principal component analysis (Kirschvink, 1980). Site-mean and over all-mean directions were calculated using Fisher statistics (Fisher, 1953). All the interpretations and paleomagnetic data processing were done using computer program packages following Enkin (1990) and Cogné (2003).

4. Paleomagnetic results

4.1. New observations

In the Luanping section, 84 core samples of 12 sites were collected from andesite beds (LP01–06 and LP08–LP13, Table 1). The natural remanent magnetization (NRM) intensities ranged 10–500 mA/m. The representative Zijderveld plots (after Zijderveld, 1967) are shown in Fig. 4a and b. A low temperature component (LTC) could be identified in most specimens between room temperature and 300 °C. The directions of the LTC in situ cluster around that of the local modern geomagnetic field direction (D = −7.0°, I = 59.6°, IGRF online data, Fig. 5). We thus consider that the LTC is a viscous remanent magnetization (VRM) of the recent magnetic field. After removing the LTC, vectors of a stable high temperature component (HTC) decayed toward the origin of coordinate near the unblock temperatures of 580 °C or 645 °C. The HTC directs NE and down. Only one-polarity directions were observed from this section. The site-mean direction is Dg = 68.3°, Ig = 66.1°, k = 77.6, and α_{95} = 5.0° in situ, which is significantly different from the modern field direction, and Ds = 30.0°, Is = 64.9°, k = 77.6, and α_{95} = 5.0° after tilt correction (Fig. 6).

In the Beipiao basin, we sampled 89 core samples of 11 sites from gray andesites (BP01–BP11, Table 1). NRM intensities ranged 20–600 mA/m (Fig. 4c and d). In most cases, an LTC could be defined below 275 °C. The directions of the LTC in situ are clustered around the local modern geomagnetic field direction (D = −8.2°, I = 60.2°, IGRF online data) and are regarded as a VRM of the recent magnetic field (Fig. 5). The HTC was defined between 375 °C and unblock temperatures (580 or 680 °C). It directs SW and up. Only one-polarity was observed. The in situ site-mean direction of the HTC is Dg = 302.5°, Ig = −68.4°, k = 29.6, and α_{95} = 8.5°, which is obviously different from that of the local geomagnetic field. After tilt correction, the site-mean directions of section I (sites BP01–06) and section II (sites BP07–11) are Ds = 183.8°, Is = −58.8°, k = 326.7, α_{95} = 3.7° and Ds = 219.5°, Is = −70.4°, k = 297.5, and α_{95} = 4.4°, respectively (Fig. 6).

In the Shouwangfen section, 21 core samples of 3 sites were collected from andesite flows (SWF12–14, Table 1). The NRM intensities ranged 100–900 mA/m; the representative Zijderveld plots are shown in Fig. 4e and f. An LTC can be identified in most specimens between room temperature and 400 °C. The directions of the LTC in situ cluster around that of the local modern geomagnetic field direction (D = −6.7°, I = 59.0°, IGRF online data, Fig. 5), representing a VRM of the recent magnetic field. After removing the LTC, vectors of a stable HTC decayed toward the origin of coordinate at unblock temperatures of 580 °C or 680 °C. The HTC directs NE and down (Fig. 6). The site-mean direction is Dg = 329.6°, Ig = 70.5°, k = 45.5, and α_{95} = 18.5° in situ, and Ds = 41.5°, Is = 48.6°, k = 42.0, and α_{95} = 19.3° after tilt correction.

4.2. Analysis of the paleosecular variation (PSV)

We list paleomagnetic data of all sites in Table 1, including those of Pei et al. (2011) for comparison. It is obvious that there is significant difference in direction among the sampling locations of the Tiaojishan Fm in both declination and inclination (Fig. 6). Our careful field geological observations indicate no tectonic reason that could be responsible for the discrepancy. In particular, no systemic difference of the data can be found between the eastern and western sampled areas. Because (1) the paleomagnetic records of each volcanic flow may provide only spot readings of geomagnetic field behavior (Tauxe, 1993), and (2) the paleosecular variation (PSV) can at any time and location easily result in a deviation of tens of degrees in geomagnetic field direction (Deenen et al., 2011), it is necessary to examine the role of PSV in the available paleomagnetic data of the Tiaojishan Fm.

Table 1
Site-mean values and statistical for the Tiaojishan Formation (~155 Ma) volcanic rocks in the Luanping, Beipiao, Shouwangfen, and Chengde areas, North China.

Site	n/N	Coor.	Direction			α_{95} (°)	Paleopole				
			D (°)	I (°)	k		Plat (°N)	Plon (°E)	dp (°)	dm (°)	Strike/dip
Luanping section (40.85°N, 117.44°E)											
LP01	3/5	G	71.0	53.2	42.5	19.1	34.6	190.2	26.5	18.4	233/17
		S	47.2	55.1	42.5	19.1	53.0	199.9	27.1	19.3	
LP02	9/9	G	85.4	63.8	135.5	4.4	30.6	171.7	7.0	5.6	233/17
		S	47.5	68.0	135.5	4.4	56.1	173.6	7.4	6.2	
LP03	9/10	G	79.1	64.5	88.7	5.5	34.9	173.2	7.1	8.8	233/17
		S	41.0	66.7	88.7	5.5	60.2	177.1	7.5	9.1	
LP04	8/8	G	78.7	61.8	31.0	10.1	33.7	177.0	12.1	15.6	233/17
		S	44.7	64.5	31.0	10.1	57.6	182.4	13.0	16.2	
LP05	7/7	G	55.9	62.4	73.7	7.1	50.6	175.1	9.7	11.8	233/17
		S	20.3	70.9	73.7	7.1	70.4	153.4	12.4	10.7	
LP06	6/6	G	11.1	70.9	84.8	7.3	73.8	140.6	11.1	12.7	233/17
		S	349.7	57.1	84.8	7.3	81.4	8.8	7.7	10.6	
LP08	6/6	G	61.3	68.8	25.1	13.6	47.7	170.4	23.1	19.5	233/17
		S	21.0	65.1	25.1	13.6	73.7	177.9	22.0	17.8	
LP09	4/4	G	52.7	66.5	57.0	12.3	52.6	176.7	20.2	16.7	233/17
		S	18.9	61.2	57.0	12.3	75.8	195.5	18.9	14.5	
LP10	5/7	G	80.9	62.9	149.7	6.3	32.9	174.7	9.9	7.8	233/17
		S	45.0	66.0	149.7	6.3	57.6	178.8	10.3	8.4	
LP11	5/6	G	69.3	60.8	18.6	18.2	39.5	182.0	27.8	21.1	233/17
		S	38.2	61.0	18.6	18.2	61.6	192.3	27.9	21.3	
LP12	6/8	G	80.1	67.6	71.9	8.0	36.0	168.2	13.4	11.1	233/17
		S	35.8	69.2	71.9	8.0	63.2	169.0	13.6	11.6	
LP13	5/8	G	67.7	73.3	28.3	14.6	45.2	159.9	26.1	23.4	233/17
		S	15.9	69.9	28.3	14.6	73.6	152.7	25.0	21.4	
Mean	73/84	G	68.3	66.1	77.6	5.0					
		S	30.0	64.9	77.6	5.0					
Beipiao section I (41.81°N, 120.60°E)											
BP01	5/8	G	264.3	−72.3	136.6	6.6	−37.0	342.7	10.3	11.7	253/34
		S	194.7	−55.4	136.6	6.6	−77.2	53.0	6.7	9.4	
BP02	9/10	G	279.5	−80.4	100.4	5.2	−36.3	323.7	9.6	10.0	253/34
		S	180.0	−59.2	100.4	5.2	−88.2	120.6	5.8	7.8	
BP03	4/7	G	282.8	−75.0	158.0	6.1	−30.7	333.0	10.2	11.3	253/34
		S	190.4	−60.8	158.0	6.1	−82.3	27.1	7.1	9.3	
BP04	9/10	G	295.1	−83.1	96.4	5.3	−35.0	315.7	10.2	10.4	253/34
		S	173.3	−60.2	96.4	5.3	−84.9	204.9	6.1	8.0	
BP05	5/6	G	277.0	−80.0	235.4	5.0	−36.8	324.9	9.2	9.6	253/34
		S	180.9	−58.9	235.4	5.0	−87.7	102.8	5.6	6.7	
BP06	6/7	G	268.1	−79.4	200.3	4.7	−39.3	327.5	8.5	9.0	253/34
		S	182.2	−57.3	200.3	4.7	−85.8	96.5	5.0	6.9	
Mean	38/48	G	275.8	−78.5	326.7	3.7					
		S	183.8	−58.8	326.7	3.7					
Beipiao section II (41.83°N, 120.71°E)											
BP07	8/9	G	314.1	−46.9	76.9	6.4	8.2	340.5	5.3	8.3	252/43
		S	242.1	−71.0	76.9	6.4	−48.3	349.6	9.7	11.1	
BP08	9/12	G	314.9	−56.9	823.9	1.8	0.7	334.9	1.9	2.6	252/43
		S	210.9	−70.7	823.9	1.8	−65.9	347.0	2.7	3.1	
BP09	6/8	G	308.5	−58.1	138.3	5.7	−3.2	338.4	6.2	8.4	252/43
		S	210.6	−67.1	138.3	5.7	−67.4	359.5	7.8	9.5	
BP10	5/6	G	315.0	−55.6	203.7	5.4	1.8	335.6	5.5	7.7	252/43
		S	214.9	−71.2	203.7	5.4	−63.5	347.0	8.2	9.4	
BP11	5/6	G	312.3	−53.7	131.4	6.7	2.3	338.4	6.5	9.4	252/43
		S	221.3	−70.1	131.4	6.7	−60.3	352.0	9.9	11.5	
Mean	33/41	G	313.0	−54.3	300.9	4.4					
		S	219.5	−70.4	297.5	4.4					
Shouwangfen section (40.598°N, 117.760°E)											
SWF12	5/5	G	345.4	60.9	68.5	9.3	79.0	39.5	14.2	10.9	71/42
		S	32.7	38.7	68.5	9.3	56.6	232.1	11.1	6.6	
SWF13	8/8	G	291.0	73.6	23.8	11.6	44.3	73.6	20.8	18.7	71/42
		S	49.9	60.4	23.8	11.6	52.9	189.8	17.6	13.4	
SWF14	8/8	G	338.0	72.6	21.2	12.3	67.7	86.2	21.8	19.4	71/42
		S	45.5	46.1	21.2	12.3	50.6	212.7	15.7	10.1	
Mean	21/21	G	329.6	70.5	45.5	18.5					
		S	41.5	48.6	42.0	19.3					
Chengde section (Pei et al., 2011) (40.8°N, 118.1°E)											
CT38	10/10	G	250.7	−39.7	6.9	19.8	−28.8	22.2	23.8	14.3	241/61
		S	207.8	−25.1	6.9	19.8	−53.2	68.8	21.3	11.4	
CD3	7/7	G	268.1	−52.2	73.1	7.1	−22.0	3.1	9.7	6.7	241/61
		S	195.4	−39.1	73.1	7.1	−67.2	78.6	8.5	5.1	
CD4	7/7	G	292.2	−50.9	36.8	10.1	−5.7	350.5	13.6	9.2	241/61

(continued on next page)

Table 1 (*continued*)

Site	n/N	Coor.	Direction			α_{95} (°)	Paleopole				
			D (°)	I (°)	k		Plat (°N)	Plon (°E)	dp (°)	dm (°)	Strike/dip
		S	192.9	−53.9	36.8	10.1	−78.0	55.9	14.1	9.9	
CD5	8/8	G	272.0	−40.5	24.9	11.3	−13.4	9.0	13.7	8.3	241/61
		S	210.9	−41.1	24.9	11.3	−58.9	52.3	13.7	8.4	
CD6-7	7/7	G	269.4	−45.6	75.4	7.0	−17.7	7.3	8.9	5.7	241/61
		S	205.4	−38.6	75.4	7.0	−61.3	62.0	8.3	4.9	
CD8	5/5	G	267.3	−26.1	76.2	8.8	−11.0	19.3	9.5	5.1	241/61
		S	229.0	−32.5	76.2	8.8	−42.2	42.1	9.9	5.6	
CD9-10	7/7	G	253.5	−42.2	26.9	11.9	−27.8	18.8	14.6	9.0	241/61
		S	205.8	−27.7	26.9	11.9	−55.6	69.9	13.0	7.1	
CD12-13	7/8	G	94.7	44.8	79.0	6.8	13.6	184.8	8.6	5.4	241/61
		S	26.6	42.4	79.0	6.8	62.5	236.2	8.4	5.2	
CD14-15	8/8	G	86.3	44.5	51.2	7.8	19.4	189.8	9.8	6.2	241/61
		S	25.2	37.3	51.2	7.8	60.8	243.5	9.2	5.4	
Mean	66/67	G	267.7	−43.5	50.5	7.3					
		S	207.1	−37.9	48.2	7.5					
Mean all [a]	231/261	G					32.9	168.1	$K = 7.7$; $A_{95} = 9.4°$		
	N = 35	S					69.6	203.0	$K = 19.5$; $A_{95} = 5.6°$		

Note: Lat/Long, latitude/longitude of sampling sites; dp/dm, semi-axes of elliptical error of the pole at a probability of 95%; n/N, number of samples used to calculate mean/total samples demagnetized; G/S, in geographic/stratigraphic coordinate; Dg/Ig and Ds/Is, declination/inclination in situ and after tilt correction; k, precision parameter of directional average; α_{95}, the radius that the mean direction lies within 95% confidence; strike/dip, right hand strike/dip of the strata; K, precision parameter of VGP average.

[a] McFadden and McElhinny (1990) reversal test, angle between the two averages $\gamma = 7.0° < \gamma_{critical} = 9.8°$ indicates a B class result.

Data of Pei et al. (2011) and our observations indicate that the Tiaojishan volcanic strata captured at least one geomagnetic field reversal. Results from the Chengde section (Pei et al., 2011) and Shouwangfen section demonstrate that the uppermost part of the Tiaojishan Fm was deposited (and cooled down) during a normal geomagnetic polarity chron, whereas the middle-lower part of the Tiaojishan Fm was formed during a reverse geomagnetic polarity chron. Considering the tight isotopic age constraints, we suggest that the two Beipiao sections can be correlated to the middle-lower part of the Tiaojishan Fm and the reverse polarity chron, whereas the Luanping section is correlated with the uppermost part of the Tiaojishan Fm and the normal polarity chron. We carried out the PSV analysis for the two polarity groups separately.

We adopted the generally accepted approach to define PSV, which is via the VGP scatter, called *S* parameter and defined as:

$$S_T = \sqrt{(N-1)^{-1}\sum_{i=1}^{N}\Delta_i^2}(i = 1, \ldots, N) \quad (1)$$

where Δ_i is the angle between the *i*th VGP and the mean VGP (Cox, 1970); *N* is the number of sites, which is generally more than five (Butler, 1992). The total angular dispersion (S_T) is caused partially by geomagnetic SV (S_B) and partially by random errors related to the sampling and measuring process (S_W). This angular dispersion of VGPs from *N* units due to SV (S_B) is calculated as per McFadden et al. (1991):

$$S_B = \sqrt{S_T^2 - \frac{(S_W)^2}{n_{avg}}} \quad (2)$$

where S_W uses the known estimate of the precision parameters (*k* and *K*), and its paleolatitude (λ) (Cox, 1970): $S_W = \frac{81}{\sqrt{K}}$; $K = k(\frac{1}{8}(5 + 18\sin^2\lambda + 9\sin^4\lambda))^{-1}$; and n_{avg} is the number of samples per site. In this paper, we calculated VGP scatters by Eq. (2). It is 9.9 for the Chengde basin section sampled by Pei et al. (2011), and 11.3, 5.0, 6.4, and 9.9 for the Luanping basin section, Beipiao basin section I and section II, and Shouwangfen section, respectively (Table 2). By comparing these VGP scatters with the predicted dispersions calculated from the "Model G" of McFadden et al. (1991), they are all smaller than the expected *S* values for the 110–195 Ma interval (Table 2), indicating that each section alone has not averaged out the PSV.

Recently, Deenen et al. (2011, 2014) proposed a model of "additional statistical reliability envelope" to test whether the obtained statistical parameters can be explained by PSV, defined by the range of A_{95} values as a reliability envelope with a lower limit ($A_{95min} = 12 \times N^{-0.40}$) and an upper limit ($A_{95max} = 82 \times N^{-0.63}$). It depends on the (sufficient) number of samples taken, which is different from the widely accepted approach of VGP scatter analysis. A_{95} values within the envelope can be explained straightforwardly by PSV alone, whereas values below A_{95min} likely under-represent PSV, and those above A_{95max} contain an additional source of scatter aside from PSV. We obtained A_{95} at 3.8 for VGPs of normal polarity samples (from Luanping, Shouwangfen sections, and two sites of Chengde section) with $A_{95\ max}/A_{95\ min}$ at 4.3°/1.8° (N = 109). We also obtained A_{95} at 3.8 for VGPs of reverse polarity samples (from Beipiao sections and seven sites of Chengde section) with A_{95max}/A_{95min} at 4.0°/1.8° (N = 122). The A_{95} values of both normal and reverse polarity groups fall within the reliability envelope, indicating that the discrepancy in direction in all the groups is resulted from PSV rather than tectonic rotation.

Therefore, we consider to use all the data from multiple sections listed in Table 1 to average out the PSV. In practice, we get the VGP scatters of 16.4 for 17 normal polarity sites and 18.7 for 18 reversed polarity sites, both being consistent with the predicted values of the "Model G" of McFadden et al. (1991; Table 2), indicating that the current data, of either normal or reverse polarity group, can pass the PSV test.

4.3. Reversal and untilting tests

The paleomagnetic directions of the two polarity groups jointly passed a reversal test (McFadden and McElhinny, 1990) at the 95% probability level (B class) with an angle between two averages $\gamma = 7.0° < \gamma_{critical} = 9.8°$ (Table 1). Because the paleomagnetic directions of each section are biased by PSV, a fold test sensu stricto cannot be performed for the available data. Nevertheless, the precision parameters "k" of directional average and "K" of VGP average become much better after regional tilt correction (Table 1, Fig. 6), suggesting strongly that the HTC defined from the Tiaojishan Fm was acquired before tilting.

Based on the PSV analyses and the field tests, we recommend conclusively the mean pole at 69.6°N, 203.0°E ($A_{95} = 5.6°$) by averaging all the VGPs obtained from the five sections for the ~155 Ma Tiaojishan Fm (Table 3).

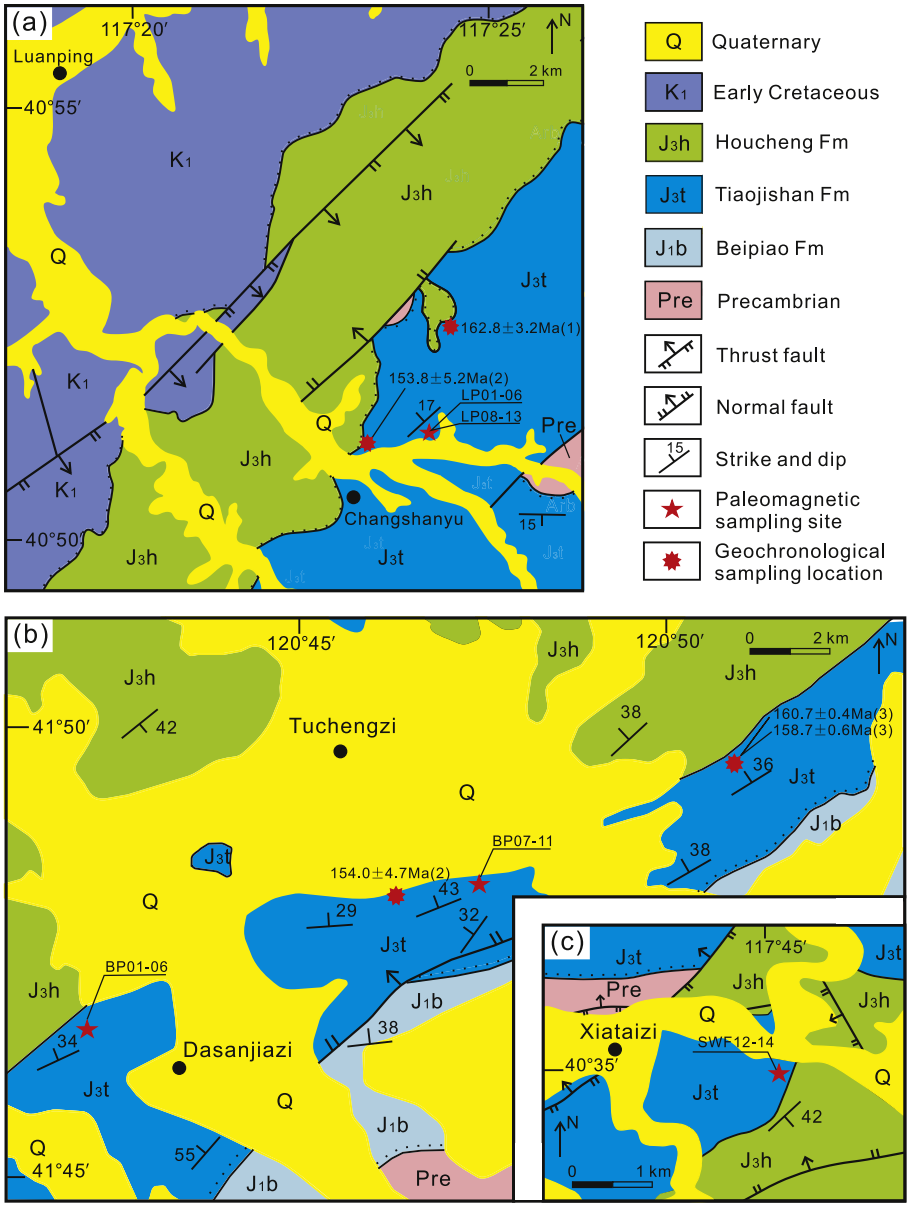


Fig. 3. Simplified geological map for sampling sections in the YSB. (a) Luanping area, (b) Beipiao area, (c) Shouwangfen area.
Modified from the Bureau of Geology and Mineral Resources of Hebei Province, 1989; Bureau of Geology and Mineral Resources of Liaoning Province, 1989. Isotopic ages are from (1) Zhang et al. (2005), (2) Zhang et al. (2008a), (3) Chang et al. (2009).

5. Discussion

5.1. Analysis of paleopoles of major blocks surrounding the NCB

Our new averaged pole of the Tiaojishan Fm (Table 3) is in good agreement with previously published Late Jurassic poles from the NCB (Fig. 7), including those by (1) Gilder and Courtillot (1997) that were obtained from the red beds and andesite of the Maotanchang and Heishidu formations near Huoshan city in the southern margin of the NCB, and by (2) Gilder et al. (1999) from Santai Formation red beds in Shandong Province in the southeastern margin of the NCB. To avoid unnecessary redundancy, some less reliable data that had been reviewed by Gilder and Courtillot (1997) and Yang et al. (1998) are not discussed in this paper. We combined all the aforementioned high-quality results by averaging their site-level VGPs to get a new pole for the Late Jurassic NCB, which is at 71.9°N, 208.6°E (A_{95} = 3.8°; Table 3). This pole is almost identical to the ~155 Ma mean pole of the Tiaojishan Fm (Table 1).

The Late Jurassic pole of the NCB is also indistinguishable from the Late Jurassic poles of the South China Block (SCB; e.g., Enkin et al., 1992; Gilder and Courtillot, 1997; Zhu et al., 1998; Yang and Besse, 2001; Yokoyama et al., 2001; Pei et al., 2011; our Table 3). This consistency can be interpreted as evidence of NCB–SCB unification amalgamation by the Late Jurassic.

Geological and geophysical evidence may demonstrate that the Mongolia block (MOB) and the NCB had merged into a single entity ("MOB–NCB") by the Late Permian or Triassic (Xiao et al., 2003; Xu et al., 2013; Zhao et al., 2013; Zhang et al., 2014a). However, the paleomagnetic results presented here for the Tiaojishan Fm are in serious conflict with some of those obtained from the MOB (Zhao et al., 1990; Gilder and Courtillot, 1997; Gilder et al., 1999; Fig. 8). For example, some paleolatitude data (Table 3) would indicate that the MOB was located south of the NCB in the Late Jurassic, which is geographically and geologically impossible. This disagreement may be the consequence of poor age-constraint for some poles of the MOB. For example, the andesite and tuff in Manzhouli, Hua'an, from which paleomagnetic results

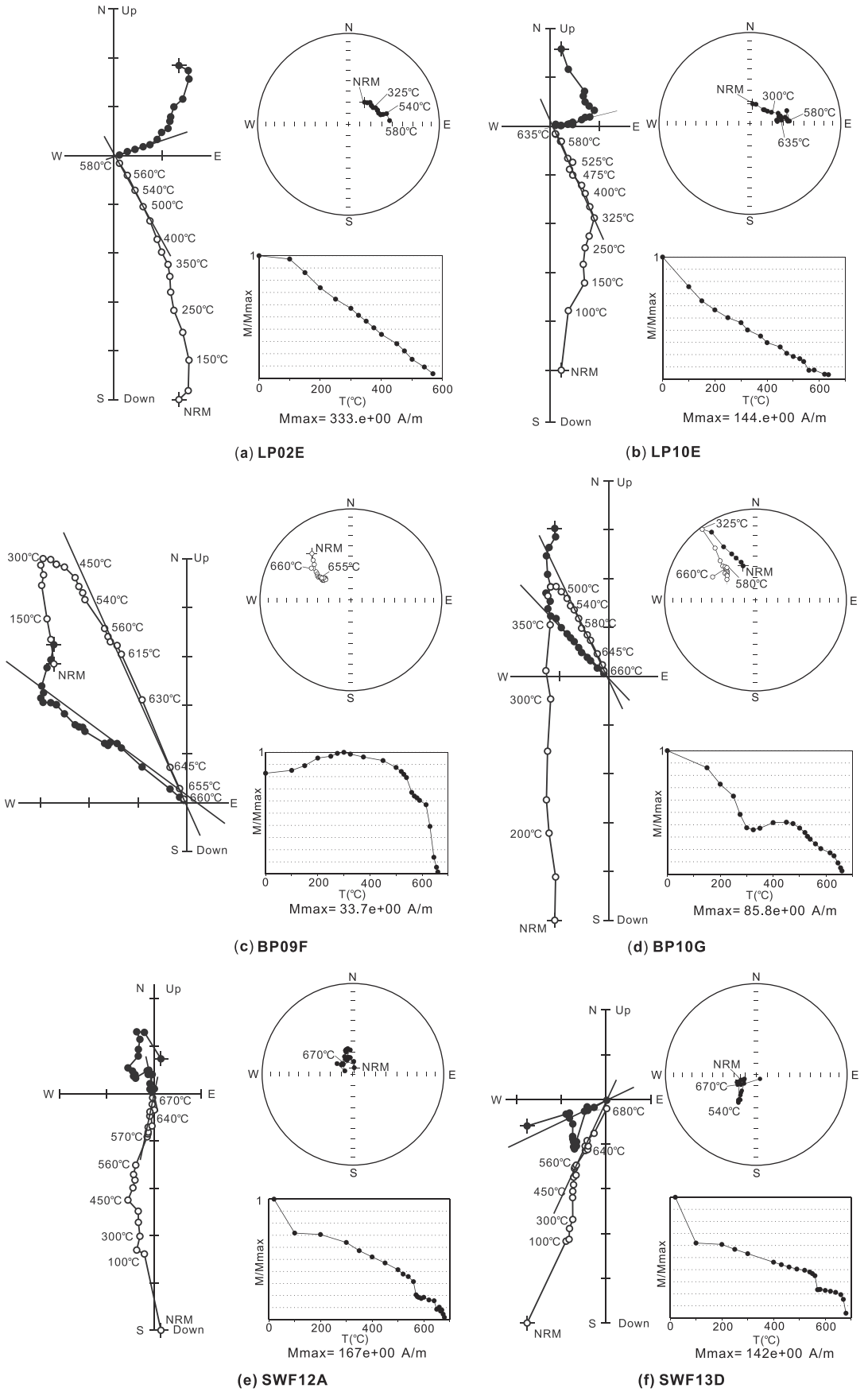


Fig. 4. Demagnetization characteristics of samples. Solid/open symbols of the orthogonal plots represent the projections onto the horizontal/vertical plane. Solid/open symbols on the stereographic projection represent the lower/upper hemisphere directions.

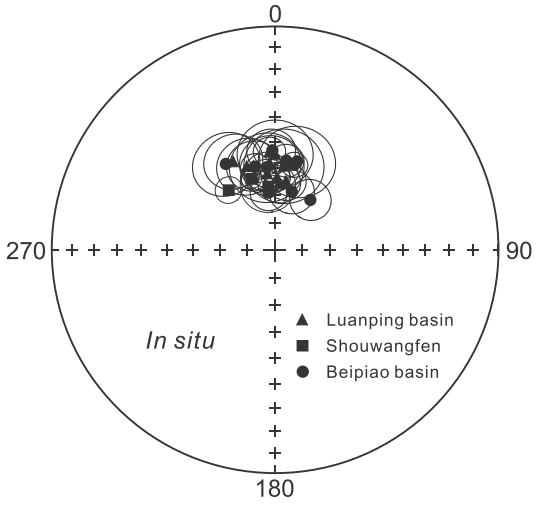


Fig. 5. Equal-area stereographic projections of the site-mean directions of the low temperature component.

were yielded, may have an age range from ~139 to ~199 Ma (Zhao et al., 1990). Data derived from the MOB Tergen and Shadaron Formations may have suffered from the same problem, because the lithological units could be either Middle Jurassic or Late Jurassic (Kravchinsky et al., 2002; Cogné et al., 2005). In fact, these MOB poles lie closer to the Middle Jurassic pole of the NCB (Fig. 8) than the Tiaojishan averaged pole, supporting the hypothesis that they are probably significantly older than Late Jurassic. Clearly, more geochronological and paleomagnetic studies are demanded (e.g., stronger field tests for some aforementioned poles) to refine the datasets of the MOB. In the following discussion, we consider the MOB and the NCB as an amalgamated tectonic unit at sometime after the Permian based on previous works (e.g., Li et al, 2012; Zhao et al., 2013; Zhang et al., 2014a). However, using the most reliable data selected, we find that the Late Jurassic paleomagnetic poles of the NCB are not consistent with coeval poles from stable Europe (Besse and Courtillot, 2002), nor with those from the Siberia craton (Kravchinsky et al., 2002; Metelkin et al., 2007a, 2010), Tarim block (Li et al., 1988), Qaidam (Halim et al., 2003), and the Jiamusi terrane (Zhang and Yang, 1996). The conclusion is unavoidable that considerable plate movement involving these three plate domains has taken place since the Late Jurassic.

Table 2
VGP scatters of the Tiaojishan Fm (~155 Ma) for PSV analysis.

Sampled section	S (°)	Palat (°N)	Expected S range of G model
Chengde*	9.9	21.7	16.6–21.0
Luanping	11.3	46.9	17.8–23.7
Shouwangfen	9.9	30.0	16.5–21.1
Beipiao I	5.0	39.6	16.5–21.1
Beipiao II	6.4	54.6	17.8–23.7
Normal polarity group	16.4	39.9	16.5–21.1
Reversed polarity group	18.7	39.6	16.5–21.1

Note: S, the VGP scatter that is calculated by Eq. (2); Palat, paleolatitude; the expected S range of G model is cited by McFadden et al. (1991); *, calculated from Pei et al. (2011) the data.

5.2. Tectonic implications

In order to integrate current paleomagnetic and geological knowledge, we present a series of paleogeographic reconstruction for the tectonic elements of Eurasia, including the Siberia craton, the Tarim, Qaidam, SCB, NCB, MOB blocks, and the Jiamusi terrane from the Late Jurassic to the Late Cretaceous (Fig. 9; Table 4). Our reconstructions were performed in a way different from previous ones by applying tighter kinematic constraints to the closure of the Mongol–Okhotsk Ocean, and to the paleogeographic positions of smaller tectonic elements surrounding the NCB.

The result of our reconstruction indicates clearly that convergence between the Siberia and NCB–MOB had not ended by ~155 Ma (Fig. 9c). The gap between them contains, most likely, the Mongol–Okhotsk Ocean basin. Reconstruction of the Late Jurassic to Early Cretaceous (Fig. 9b) shows that there are no significant variations in paleolatitude and paleoazimuth for the NCB–MOB; however, Europe and Siberia experienced a southward displacement with a slight tectonic rotation among them during the period (Fig. 9c and b). We consider that this plate movement was likely the result of the final closure of the Mongol–Okhotsk Ocean. Although we cannot directly equate paleolatitudinal differences with the width of that ocean, we do conclude that there has been ~1600 km of lithospheric shortening between Siberia and the NCB after ~155 Ma. This conclusion agrees with the analysis of this issue by Metelkin et al. (2010), but differs from that of Pei et al. (2011) who suggested ~3000 km of post-Late Jurassic crustal shortening between Siberia and NCB based on the data of the Tiaojishan Fm in Chengde section. Our result reinforces a tighter spatial-temporal constraint by matching the ~155 Ma coeval poles from two sides of the Mongol–Okhotsk Ocean.

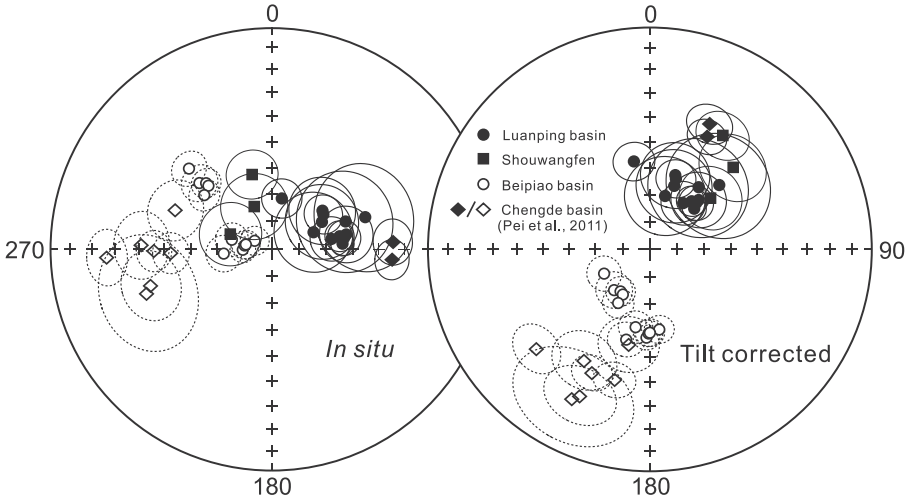


Fig. 6. Equal-area stereographic projections of the site-mean directions of the high temperature component: Lower (upper) hemisphere directions are represented by solid (open) symbols.

Table 3
The Mesozoic paleomagnetic poles from the NCB and the surrounding blocks.

Block	Age (Ma)	N	Plat (°N)	Plon (°E)	A_{95} (°)	Criterion (Q)	References
Late Cretaceous							
EUR	75		81.3	188.6	7.2		Besse and Courtillot (2002)
JMS	K_2	5	78.5	95.7	14.7	123-5-7(5)	Wang et al. (2011a)
NCB-mean *	K_2	4S	81.1	194.0	11.2		Zhao et al. (1996)
SCB-mean	K_2	11S	75.2	210.7	7.5		Lin et al. (2003)
SIB	75	16	82.2	188.5	6.1	123C5R7(7)	Metelkin et al. (2007b)
Early Cretaceous							
EUR	120		78.2	189.4	2.4		Besse and Courtillot (2002)
JMS	K_{1-2}	10	72.2	73.7	5.9	123-5-7(5)	Zhang and Yang (1996)
NCB-mean *	K_1	5S	81.7	206.8	6.7		Lin et al. (2003)
QDM	K_1	19	76.0	187.2	5.4	123F5-7(6)	Sun et al. (2006)
SCB-mean	K_1	8S	78.8	202.1	7.1		Lin et al. (2003)
SIB	120	25	72.3	186.4	6.0	123F5R7(7)	Metelkin et al. (2004)
TAR	115	13	64.1	172.1	12.0	123F5R7(7)	Gilder et al. (2003)
Late Jurassic							
EUR	150		75.0	159.9	6.6		Besse and Courtillot (2002)
JMS [1]	J_3	3	71.0	63.4	9.9	123-5-7(5)	Zhang and Yang (1996)
NCB	155	9	59.9	240.3	6.8	123-5R7(6)	Pei et al. (2011)
NCB	155	26	70.1	184.9	5.8		
NCB-mean *	155	35	69.6	203.0	5.6	123-5R7(6)	This study
NCB	J_3	10	74.4	222.8	5.9	123F5R7(7)	Gilder and Courtillot (1997)
NCB	J_3	5	77.8	235.9	6.9	123-5R7(6)	Gilder et al. (1999)
NCB	J_3	6	73.5	207.8	5.7	123-5R7(6)	Gilder et al. (1999)
NCB-mean	J_3	5S	71.9	208.6	3.8		
QDM [2]	J_3	9	50.1	198.0	6.6	123F5-7(6)	Halim et al. (2003)
SCB-mean [3]	J_3	5S	70.4	220.9	8.5		Pei et al. (2011)
SIB [4]	155	9	64.4	161.0	7.0	123F5-7(6)	Kravchinsky et al. (2002)
SIB [5]	155	18	63.6	166.8	8.5	123F5R7(7)	Metelkin et al. (2007a)
TAR [6]	J_3	6	64.6	208.9	9.0	123-5R7(6)	Li et al. (1988)
Middle Jurassic–Late Triassic							
Inner Mongolia [7]	J	8	73.0	254.8	7.8	−23-5R7(5)	Zhao et al. (1990)
Inner Mongolia [8]	J	14	62.4	224.6	4.9	−23F5R7(6)	Zhao et al. (1990)
MOB [9]	J_{2-3}	6	73.3	275.9	6.3	123-5-7(5)	Cogné et al. (2005)
NCB	J_2	6	74.3	232.8	5.0	123-5R7(6)	Yang et al. (1992)
NCB	J_2	17	72.9	254.7	6.4	123F5R7(7)	Gilder and Courtillot (1997)
NCB-mean *	J_2	23	73.6	249.3	4.9		Gilder and Courtillot (1997)
NCB *	J_1	10	82.4	286.0	6.8	123-5R7(6)	Yang et al. (1998)
NCB *	T_3	11	62.3	7.7	3.8	123-5R7(6)	Yang et al. (1998)

Note: Ages of the rock units [T_3, Upper Triassic; J_1, Lower Jurassic; J_2, Middle Jurassic; J_3, Upper Jurassic (includes 150–160 Ma); K_1, Lower Cretaceous; K_2, Upper Cretaceous]; N, number of sites or Studies (S) used for pole determination; Plat and Plon, latitude and longitude of the pole; A_{95}, the radius that the mean pole lies within 95% confidence; criterion (Q, numbers of the criteria met), reliability criteria 1–7 from Van der Voo (1990) [1, well-dated rocks; 2, sufficient numbers of samples, n > 25, k > 10 and $\alpha_{95} < 16°$; 3, details of demagnetization; 4, field test (F, the positive fold test; C, positive baked-contact test); 5, tectonic coherence with the craton and structural control; 6, presence of reversals; 7, no similarity to younger paleopoles in the same craton]; "–", failed to meet this criterion. [1]–[9], used for Fig. 8. *, used to refine the APWP of the NCB in Fig. 8. Underlined data are used for reconstructions of Fig. 9. For abbreviations see caption of Fig. 1a.

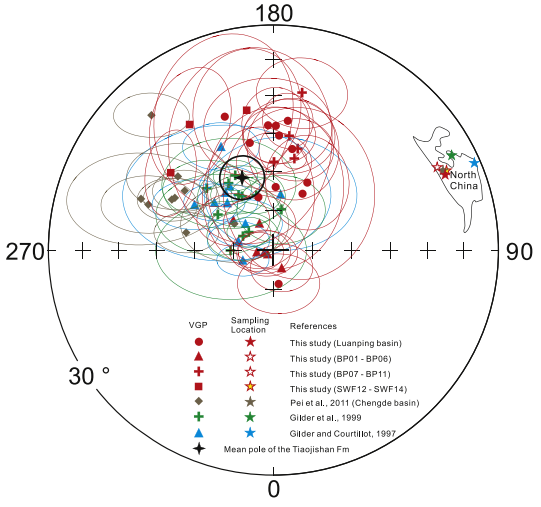


Fig. 7. Equal-area projection (>30°N region only) showing the site-level virtual geomagnetic poles (VGPs) available of the ~155 Ma Tiaojishan Fm and Late Jurassic successions of the NCB. Ellipses of 95% confidence are drawn around each VGP.

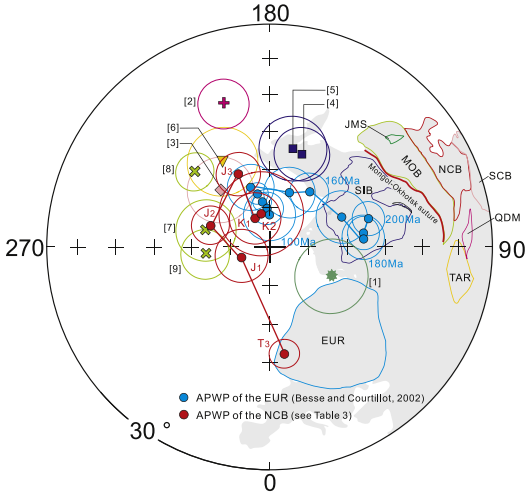


Fig. 8. Equal-area projection (>30°N region only) showing the Mesozoic segments of the APWPs for the NCB and Eurasia, and the Late Jurassic poles for the surrounding blocks (listed in Table 3). For abbreviations see caption of Fig. 1a.

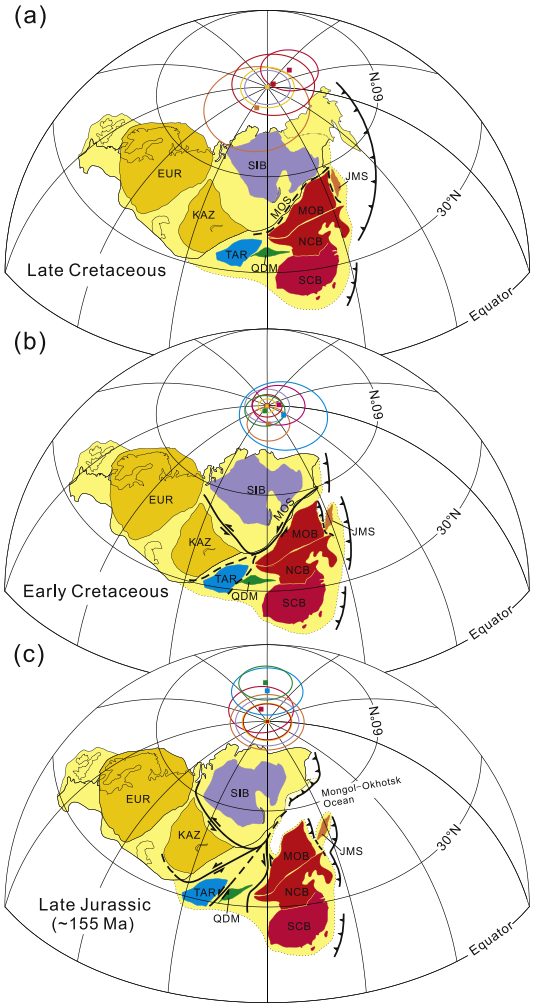


Fig. 9. Schematic reconstructions of the major Eurasia blocks from the Late Jurassic (~155 Ma) to Late Cretaceous using the Euler rotation parameters (listed in Table 4). For abbreviations see caption of Fig. 1a.

Table 4
The paleopoles and Euler rotation parameters for Late Jurassic to Late Cretaceous reconstructions.

Block	Euler rotation parameters
Late Jurassic	
EUR	(0, 69.9, 15.0)
JMS	(61.6, 135.4, −40.6)
NCB–MOB	(0. 113.0, 20.4)
QDM	(16.8, 90.7, 38.0)
SCB	(0, 113.0, 20.4)
SIB	(36.5, 86.8, 33.0)
TAR	(7.7, 89.1, 26.5)
Early Cretaceous	
EUR	(0, 99.4, 11.8)
JMS	(63.6, 129.4, −33.1)
NCB–MOB	(35.8, 293.8, −10.2)
QDM	(8.6, 104.6, 15.1)
SCB	(35.8, 293.8, −10.2)
SIB	(8.0, 100.4, 177.7)
TAR	(18.4, 97.1, 20.5)
Late Cretaceous	
EUR	(0, 98.6, 8.7)
JMS	(45.0, 217.7, −7.6)
NCB–MOB	(10.0, 276.5, −7.3)
SCB	(10.0, 276.5, −7.3)
SIB	(0, 98.5, 7.8)

For abbreviations see caption of Fig. 1a.

We agree with previous paleomagnetic-based hypotheses (Cogné et al., 2005; Metelkin et al., 2007a, 2010 and references therein) that a large intraplate sinistral strike–slip system developed in central Asia in the Late Jurassic. The strike–slip system appears to have accommodated displacement between the eastern Asian blocks caused by the closure-in-progress of the Mongol–Okhotsk Ocean (Fig. 9). For example, the Tarim block, as a tectonic link between Europe and eastern Asian blocks, may have been juxtaposed at the eastern margin of the Kazakhstan by Late Jurassic time (Zhao et al., 1990; Kravchinsky et al., 2002; Gilder et al., 2008). However, Late Jurassic to Early Cretaceous paleomagnetic poles for the two blocks indicate a significant latitudinal displacement between them. During the Late Jurassic and Early Cretaceous, the Tarim moved northward with respect to Kazakhstan along a sinistral strike–slip fault that might have connected with the subduction zone in the Mongol–Okhotsk Ocean along the southern margin of the Siberian plate (Fig. 9c). As an independent tectonic unit, the much smaller Jiamusi block might have been influenced by the subduction of the Pacific plate with a resultant ~30° counterclockwise rotation from Late Jurassic to Late Cretaceous time.

The convergence between Siberian plate and MOB–NCB slowed down gradually during the Early Cretaceous. The paleomagnetic poles of Late Cretaceous for the SCB, NCB, Tarim, and Qaidam blocks are indistinguishable from those of Siberia and stable Europe, indicating that Eurasia had become internally consolidated by then (Fig. 9a).

An additional finding of this study is that there are no significant crustal rotations about a relative vertical axis between the Luanping basin in the central YSB and the Beipiao basin in the eastern YSB after ~155 Ma (Fig. 1). In an earlier study, Pei et al. (2011) observed no rotation of the Chengde basin that lies in between our two study areas. Therefore, available paleomagnetic works suggest that the NCB and much of the YSB have been tectonically coherent with respect to the interior of the North China craton since the Late Jurassic.

6. Conclusion

Our new investigation demonstrates that previously published paleomagnetic data of the Tiaojishan volcanic rocks were strongly biased by the paleosecular variation (PSV), and the PSV has been averaged out by combining all the VGPs from the Tiaojishan Fm in the region. The mean pole at 69.6°N/203.0°E (A_{95} = 5.6°) passes a reversal test and regional tilting test at 95% confidence and is thus considered as a primary paleomagnetic record. This new pole of the Tiaojishan Fm is consistent with the available Late Jurassic poles from red-beds in the southern part of the NCB, but they are distinguishable from coeval poles of Siberia and the reference pole of Eurasia, indicating that convergence between the Siberia and the NCB had not ended by ~155 Ma. Our calculation indicates a ~1600-km latitudinal plate movement and crustal shortening between the Siberian plate and the NCB after ~155 Ma. In addition, no significant vertical axis rotation was found either between our sampled areas within the Yanshan Belt or between the Yanshan Belt and the major part of the NCB after ~155 Ma.

Acknowledgments

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References

Besse, J., Courtillot, V., 2002. Apparent and true polar wander and the geometry of the geomagnetic field over the last 200 Myr. Journal of Geophysical Research 107, 101029–101060.

Bureau of Geology and Mineral Resources of Hebei Province, 1989. Regional Geology of Hebei Province (in Chinese). Geological Publishing House, Beijing, pp. 21–351.
Bureau of Geology and Mineral Resources of Liaoning Province, 1989. Regional Geology of Liaoning Province (in Chinese). Geological Publishing House, Beijing, pp. 26–322.
Butler, R.F., 1992. Paleomagnetism: Magnetic Domains to Geologic Terranes. Blackwell, Boston, pp. 1–319.
Chang, S.-C., Zhang, H., Renne, P.R., Fang, Y., 2009. High-precision $^{40}Ar/^{39}Ar$ age constraints on the basal Lanqi Formation and its implications for the origin of angiosperm plants. Earth and Planetary Science Letters 279, 212–221.
Chen, Y., Chen, W., 1997. Mesozoic Volcanic Rocks in Liaoxi and Its Adjacent Regions: Chronology, Geochemistry and Tectonic Settings. Seismological Press, Beijing, pp. 1–279 (in Chinese with English summary).
Cogné, J.P., 2003. PaleoMac: a Macintosh™ application for treating paleomagnetic data and making plate reconstructions. Geochemistry, Geophysics, Geosystems 4 (1), 1007. http://dx.doi.org/10.1029/2001 GC000227.
Cogné, J.P., Kravchinsky, V.A., Halim, N., Hankard, F., 2005. Late Jurassic–Early Cretaceous closure of the Mongol–Okhotsk Ocean demonstrated by new Mesozoic palaeomagnetic results from the Trans-Baïkal area (SE Siberia). Geophysical Journal International 163, 813–832.
Cox, A., 1970. Latitude dependence of the angular dispersion of the geomagnetic field. Geophysical Journal International 20, 253–269.
Davis, G.A., Zheng, Y.D., Wang, C., Darby, B.J., Zhang, C.H., Gehrels, G.F., 2001. Mesozoic tectonic evolution of the Yanshan fold and thrust belt, with emphasis on Hebei and Liaoning provinces, northern China. Geological Society of America Memoirs 194, 171–197.
Davis, G.A., Meng, J.F., Cao, W.R., Du, X.Q., 2009. Triassic and Jurassic tectonics in the Eastern Yanshan Belt, North China: insights from the controversial Dengzhangzi Formation and its neighboring units. Earth Science Frontiers 16 (3), 69–86.
Deenen, M.H.L., Langereis, C.G., van Hinsbergen, D.J.J., Biggin, A.J., 2011. Geomagnetic secular variation and the statistics of palaeomagnetic directions. Geophysical Journal International 186, 509–520.
Deenen, M.H.L., Langereis, C.G., van Hinsbergen, D.J.J., Biggin, A.J., 2014. Erratum: Geomagnetic secular variation and the statistics of palaeomagnetic directions. Geophysical Journal International 197, 643.
Enkin, R.J., 1990. Formation et deformation de l'Asie depuis la fin de l'ere primaire: les apports de l'etude paleomagnetique des formations secondaires de Chine du Sud Ph D thesis. Paris University.
Enkin, R.J., Yang, Z.Y., Chen, Y., Courtillot, V., 1992. Paleomagnetic constraints on the geodynamic history of the major blocks of China from the Permian to the present. Journal of Geophysical Research 97, 13953–13989.
Fang, D.J., Guo, Y.B., Wang, Z.L., Tan, X.D., Fan, S.Q., Yuan, Y.R., Tang, X.Z., Wang, B.G., 1988. The tectonic implication of the paleomagnetic results from Triassic and Jurassic in Ningwu basin, Shanxi Province. Chinese Science Bulletin 2, 133–135.
Fisher, R., 1953. Dispersion on a sphere. Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences 217, 295–305.
Gilder, S.A., Courtillot, V., 1997. Timing of the North–South China collision from new middle to late Mesozoic paleomagnetic data from the North China Block. Journal of Geophysical Research 102, 17713–17727.
Gilder, S.A., Leloup, P.H., Courtillot, V., Chen, Y., Coe, R.S., Zhao, X., Xiao, W., Halim, N., Cogné, J.P., Zhu, R., 1999. Tectonic evolution of the Tancheng–Lujiang (Tan–Lu) fault via middle Triassic to Early Cenozoic paleomagnetic data. Journal of Geophysical Research 104, 15365–15390.
Gilder, S.A., Chen, Y., Cogné, J.-P., Tan, X., Courtillot, V., Sun, D., Li, Y., 2003. Paleomagnetism of Upper Jurassic to Lower Cretaceous volcanic and sedimentary rocks from the western Tarim Basin and implications for inclination shallowing and absolute dating of the M-0 (ISEA?) chron. Earth and Planetary Science Letters 206, 587–600.
Gilder, S.A., Gomez, J., Chen, Y., Cogné, J.P., 2008. A new paleogeographic configuration of the Eurasian landmass resolves a paleomagnetic paradox of the Tarim Basin (China). Tectonics 27, TC1012. http://dx.doi.org/10.1029/2007TC002155.
Halim, N., Kravchinsky, V., Gilder, S., Cogné, J.P., Alexyutin, M., Sorokin, A., Courtillot, V., Yan, C., 1998. A palaeomagnetic study from the Mongol–Okhotsk region: rotated Early Cretaceous volcanics and remagnetized Mesozoic sediments. Earth and Planetary Science Letters 159, 133–145.
Halim, N., Chen, Y., Cogné, J.P., 2003. A first palaeomagnetic study of Jurassic formations from the Qaidam basin, Northeastern Tibet, China — tectonic implications. Geophysical Journal International 153, 20–26.
Kirschvink, J., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophysical Journal International 62, 699–718.
Kravchinsky, V.A., Cogné, J.P., Harbert, W.P., Kuzmin, M.I., 2002. Evolution of the Mongol–Okhotsk Ocean as constrained by new palaeomagnetic data from the Mongol–Okhotsk suture zone, Siberia. Geophysical Journal International 148, 34–57.
Li, Y.P., Zhang, Z.K., McWilliams, M., Sharps, R., Zhai, Y.J., Li, Y.A., Li, Q., Cox, A., 1988. Mesozoic paleomagnetic results of the Tarim Craton: tertiary relative motion between China and Siberia? Geophysical Research Letters 15, 217–220.
Li, P.W., Zhang, S.H., Gao, R., Li, H.Y., Zhao, Q.L., Li, Q.S., Guan, Y., 2012. New Upper Carboniferous–Lower Permian paleomagnetic results from the central inner Mongolia and their geological implications. Journal of Jilin University (Earth Science Edition) 42 (Sup. 1), 423–440.
Lin, J.L., Fuller, M., Zhang, W.Y., 1985. Preliminary Phanerozoic polar wander paths for the North and South China blocks. Nature 313 (7), 444–449.
Lin, W., Chen, Y., Faure, M., Wang, Q., 2003. Tectonic implications of new Late Cretaceous paleomagnetic constraints from eastern Liaoning Peninsula, NE China. Journal of Geophysical Research, Solid Earth 108 (B6), 2313. http://dx.doi.org/10.1029/2002JB002169.
Liu, J., Zhao, Y., Liu, X., 2006. Age of the Tiaojishan Formation volcanics in the Chengde basin, northern Hebei Province. Acta Petrologica Sinica 22, 2617–2630.
Liu, S.F., Zhang, J.F., Hong, S.Y., Ritts, B.D., 2007. Early Mesozoic basin development and its response to thrusting in the Yanshan fold-and-thrust belt, China. International Geology Review 49, 1025–1049.
Maruyama, S., Isozaki, Y., Kimura, G., Terabayashi, M., 1997. Paleogeographic maps of the Japanese islands: plate tectonic synthesis from 750 Ma to the present. Island Arc 6, 121–142.
McFadden, P., McElhinny, M., 1990. Classification of the reversal test in palaeomagnetism. Geophysical Journal International 103, 725–729.
McFadden, P., Merrill, R., McElhinny, M., Lee, S., 1991. Reversals of the Earth's magnetic field and temporal variations of the dynamo families. Journal of Geophysical Research 96, 3923–3933.
Metelkin, D.V., Gordienko, I., Zhao, X., 2004. Paleomagnetism of Early Cretaceous volcanic rocks from Transbaikalia: argument for Mesozoic strike–slip motions in Central Asian structure. Russian Geology and Geophysics 45, 1404–1417.
Metelkin, D.V., Gordienko, I., Klimuk, V., 2007a. Paleomagnetism of Upper Jurassic basalts from Transbaikalia: new data on the time of closure of the Mongol–Okhotsk Ocean and Mesozoic intraplate tectonics of Central Asia. Russian Geology and Geophysics 48, 825–834.
Metelkin, D.V., Kazansky, A.Y., Bragin, V.Y., Tsel'movich, V., Lavrenchuk, A., Kungurtsev, L., 2007b. Paleomagnetism of the Late Cretaceous intrusions from the Minusa trough (southern Siberia). Russian Geology and Geophysics 48, 185–198.
Metelkin, D.V., Vernikovsky, V.A., Kazansky, A.Y., Wingate, M.T., 2010. Late Mesozoic tectonics of Central Asia based on paleomagnetic evidence. Gondwana Research 18, 400–419.
Parfenov, L., Popeko, L., Tomurtogoo, O., 2001. Problems of tectonics of the Mongol–Okhotsk orogenic belt. Geology of the Pacific Ocean 16, 797–830.
Pei, J.L., Sun, Z.M., Liu, J., Liu, J., Wang, X.S., Yang, Z.Y., Zhao, Y., Liu, H.B., 2011. A paleomagnetic study from the Late Jurassic volcanics (155 Ma), North China: implications for the width of Mongol–Okhotsk Ocean. Tectonophysics 510, 370–380.
Sun, Z.M., Yang, Z.Y., Pei, J.L., Yang, T.S., Wang, X.S., 2006. New Early Cretaceous paleomagnetic data from volcanic and red beds of the eastern Qaidam Block and its implications for tectonics of Central Asia. Earth and Planetary Science Letters 243, 268–281.
Tauxe, L., 1993. Sedimentary records of relative paleointensity of the geomagnetic field: theory and practice. Reviews of Geophysics 31, 319–354.
Tomurtogoo, O., Windley, B., Kröner, A., Badarch, G., Liu, D., 2005. Zircon age and occurrence of the Adaatsag ophiolite and Muron shear zone, central Mongolia: constraints on the evolution of the Mongol–Okhotsk ocean, suture and orogen. Journal of the Geological Society 162, 125–134.
Van der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 184, 1–9.
Wang, Y., 1996. Tectonic Evolutional Processes of Inner Mongolia–Yanshan Orogenic Belt in Eastern China During the Late Paleozoic–Mesozoic. Geological Publishing House, Beijing.
Wang, Y., Zhou, L., Zhao, L., Ji, M., Gao, H., 2010. Palaeozoic uplands and unconformity in the North China Block: constraints from zircon LA-ICP-MS dating and geochemical analysis of bauxite. Terra Nova 22, 264–273.
Wang, H.L., Huang, B.C., Qiao, Q.Q., Chen, J.S., 2011a. Paleomagnetic study on Cretaceous and Plaeogene rocks from eastern Heilongjiang, NE China and its tectonic implications. Chinese Journal of Geophysics 54 (3), 793–806 (in Chinese).
Wang, Y., Zhou, L.Y., Li, J.Y., 2011b. Intracontinental superimposed tectonics—a case study on the Western Hills of Beijing, eastern China. Geological Society of America Bulletin 123, 1033–1055.
Wang, Y., Zhou, L., Zhao, L., 2013. Cratonic reactivation and orogeny: an example from the northern margin of the North China Craton. Gondwana Research 24, 1203–1222.
Xiao, W.J., Windley, B.F., Hao, J., Zhai, M.G., 2003. Accretion leading to collision and the Permian Solonker suture, Inner Mongolia, China: termination of the central Asian orogenic belt. Tectonics 22, 1069.
Xu, X., Harbert, W., Dril, S., Kravchinksy, V., 1997. New paleomagnetic data from the Mongol–Okhotsk collision zone, Chita region, south-central Russia: implications for Paleozoic paleogeography of the Mongol–Okhotsk ocean. Tectonophysics 269, 113–129.
Xu, H., Liu, Y.Q., Kuang, H.W., Jiang, X.J., Peng, N., 2012. U–Pb SHRIMP age for the Tuchengzi Formation, northern China, and its implications for biotic evolution during the Jurassic–Cretaceous transition. Palaeoworld 21, 222–234.
Xu, B., Charvet, J., Chen, Y., Zhao, P., Shi, G.Z., 2013. Middle Paleozoic convergent orogenic belts in western Inner Mongolia (China): framework, kinematics, geochronology and implications for tectonic evolution of the Central Asian Orogenic Belt. Gondwana Research 23 (4), 1342–1364.
Yang, Z.Y., Besse, J., 2001. New Mesozoic apparent polar wander path for south China: tectonic consequences. Journal of Geophysical Research, Solid Earth 106, 8493–8520.
Yang, Z.Y., Courtillot, V., Besse, J., Ma, X.H., Xing, L.S., Xu, S.J., Zhang, J.X., 1992. Jurassic paleomagnetic constraints on the collision of the North and South China blocks. Geophysical Research Letters 19, 577–580.
Yang, Z.Y., Ma, X.H., Huang, B.C., Sun, Z.M., Zhou, Y.X., 1998. Apparent polar wander path and tectonic movement of the North China Block in Phanerozoic. Science in China Series D: Earth Sciences 41, 51–65.
Yokoyama, M., Liu, Y., Halim, N., Otofuji, Y.-i., 2001. Paleomagnetic study of Upper Jurassic rocks from the Sichuan basin: tectonic aspects for the collision between the Yangtze Block and the North China Block. Earth and Planetary Science Letters 193, 273–285.
Zhang, S.H., Yang, H.X., 1996. Paleomagnetism of the Jiamusi terrane in the late Jurassic epoch and Cretaceous period and its tectonic significance, NE China. Journal of Changchun University of Science and Technology 2, 207–210.
Zhang, H., Yuan, H., Hu, Z., Liu, X., Diwu, C., 2005. U–Pb zircon dating of the Mesozoic volcanic strata in Luanping of North Hebei and its significance. Earth Sciences—Journal of China University of Geosciences 30, 707–720.

Zhang, H., Wang, M., Liu, X., 2008a. Constraints on the upper boundary age of the Tiaojishan Formation volcanic rocks in West Liaoning–North Hebei by LA-ICP-MS dating. Chinese Science Bulletin 53, 3574–3584.

Zhang, H., Wei, Z.L., Liu, X.M., Li, D., 2008b. U–Pb zircon dating of the Tuchengzi Formation in Jibei–Liaoxi. Science in China Series D: Earth Sciences 38, 960–970.

Zhang, S.H., Gao, R., Li, H.Y., Hou, H.S., Wu, H.C., Li, Q.S., Yang, K., Li, C., Li, W.H., Zhang, J.S., Yang, T.S., Keller, G.R., Liu, M., 2014a. Crustal structures revealed from a deep seismic reflection profile across the Solonker suture zone of the Central Asian Orogenic Belt, northern China: an integrated interpretation. Tectonophysics 612–613, 26–39. http://dx.doi.org/10.1016/j.tecto.2013.11.035.

Zhang, S.H., Zhao, Y., Davis, G.A., Ye, H., Wu, F., 2014b. Temporal and spatial variations of Mesozoic magmatism and deformation in the North China Craton: implications for lithospheric thinning and decratonization. Earth-Science Reviews 131, 49–87.

Zhao, Y., 1990. The Mesozoic orogenies and tectonic evolution of the Yanshan area. Geological Review 36, 1–13.

Zhao, X.X., Coe, R.S., Zhou, Y.X., Wu, H.R., Wang, J., 1990. New paleomagnetic results from northern China: collision and suturing with Siberia and Kazakhstan. Tectonophysics 181, 43–81.

Zhao, X.X., Coe, R.S., Gilder, S., Frost, G., 1996. Palaeomagnetic constraints on the palaeogeography of China: implications for Gondwanaland. Australian Journal of Earth Sciences 43, 643–672.

Zhao, P., Chen, Y., Xu, B., Faure, M., Shi, G., Choulet, F., 2013. Did the Paleo-Asian Ocean between North China Block and Mongolia Block exist during the late Paleozoic? First paleomagnetic evidence from central–eastern Inner Mongolia, China. Journal of Geophysical Research, Solid Earth 118, 1873–1894.

Zhu, R.X., Yang, Z.Y., Wu, H.N., Ma, X.H., Huang, B.C., Meng, Z.F., Fang, D.J., 1998. Paleomagnetic constraints on the tectonic history of the major blocks of China during the Phanerozoic. Science in China Series D: Earth Sciences 41, 1–19.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: Collinson, D., Creer, K., Runcorn, S. (Eds.), Methods in Paleomagnetism. Elsevier, Amsterdam, pp. 254–286.

Zonenshain, L.P., Kuzmin, M.I., Natapov, L.M., 1990. Geology of the USSR: a plate tectonic synthesis. Geodynamic Series 21. American Geophysical Union, Washington, DC (242 pp.).

Zorin, Y.A., 1999. Geodynamics of the western part of the Mongolia–Okhotsk collisional belt, Trans–Baikal region (Russia) and Mongolia. Tectonophysics 306, 33–56.